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A Comparative Neuroprimatological Study on the Inferior Olivary Nuclei (from the STEPHAN's Collection)*

Shozo MATANO

Department of Biological Anthropology, Faculty of Human Sciences, Osaka University

Abstract In 2 species of Tupaiidae, 17 species of prosimians, 12 species of New World monkeys, 10 species of Old World monkeys, *Hylobates, Gorilla, Pan* and human (the STEPHAN's collection), volumes of the inferior olivary nuclei (principal-IOPr, accessory-IOAc) were measured. The phylogenetical development was studied by using allometry based upon the regression line of 17 species of prosimians. Volume ratios of IOPr and IOAc to medulla oblongata were also examined.

The IOPr evolved progressively from prosimians to human, whereas the IOAc were independent of a general evolutionary trend of primates. The IOAc in great apes and human are far inferior to the average prosimian level. Among the monkeys, the arboreal quadrupedal type develops maintaining a balance between the IOPr and the IOAc in comparison with the semi-brachiation and the Old World terrestrial quadrupedal types. Putting these results together with those of the ventral pons (MATANO *et al.*, 1985b), cerebellar nuclear complex (MATANO *et al.*, 1985a) and the vestibular nuclear complex (MATANO, 1986), some considerations were made in relation with the motor system centering around the cerebellum.

Key Words Inferior olivary nuclei, Volumetric comparison, Primates, Motor system

Introduction

The inferior olivary nuclei consist of the principal nucleus, the medial accessory nucleus and the dorsal accessory nucleus. The principal nucleus receives the cortical fibers from frontal, parietal, temporal and occipital cortex and mainly projects the fibers to the opposite cerebellar hemisphere as the so called climbing fibers (CARPENTER, 1976, COURVILLE *et al.*, 1980). Therefore, this nucleus is considered to be an important intermediate station in the cerebro-cerebellar hemisphere circuit. On the other hand, the accessory nuclei mainly receive the fibers originating from midbrain periaqueduct gray and the fibers of spinal origin, and project their efferent fibers largely to cerebellar vermis (CARPENTER, 1976, COURVILLE *et al.*, 1980).

Because of the morphological characteristics of the inferior olivary nuclei, their size and form among various animal species have been noted by many researchers (*eg.* KAPPERS *et al.*, 1960;

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KUHLENBECK, 1975). Very early, TILNEY (1927) pointed out poor development of these nuclei in *Tarsius.* Among primates, a general discussion of comparison of these nuclei was made by GERHARD and OLSZEWSKI (1969). However, a detailed and systematic investigation of size comparison of the inferior olivary nuclei has not been made in primates.

Viewed from the phylogenetical and neurological standpoints related to the cerebellum, we studied volumetric comparisons among various primate species including human on the ventral pons (MATANO *et al.*, 1985b), the cerebellar nuclei (MATANO *et al.*, 1985a), and the vestibular nuclei (MATANO, 1986, BARON *et al.*, 1988). These results are all dependent on STEPHAN's valuable collection.

In the present paper, the volumes of three inferior olivary nuclei are measured in 2 species of Scandentia, 17 species of prosimians and 26 species of anthropoids including human from the STEPHAN's collection, and results will be discussed on the phylogenetical and neurological points related to the cerebellar motor system as one of series of the above-mentioned studies.

Materials and Method

Volumes of the principal inferior olivary nucleus, the medial accessory inferior olivary nucleus and the dorsal accessory, were measured in the brains of 78 individuals of Scandentia (2 species), prosimians (17 species) and anthropoids (26 species). The collection of brains consist of the frontal and sagittal serial sections, fixed in BOUIN's fluid, embedded in paraffin. And they are stained with cresyl violet for cells and by the HEIDENHAIN-WOELCKE method for fibers. The details of this collection are given by STEPHAN *et al.* (1981).

The border of three inferior olivary nuclei were traced on the tracing paper under the projection microscope from frontal sections taken at equal intervals. The areas traced were estimated by the Kontron-MOP-Modular system. Volumes were calculated by applying the formula $V = A \times D/M^2$. (V = volume in mm^3, A = total areas measured in mm^2, D = Distance of measured section in mm, M^2 = Square of linear magnification).

The conversion of serial section volumes into fresh volumes were also given by STEPHAN *et al.* (1981).

Results and Discussion

Data measured are shown in Table 1 (Scandentia and prosimians), Tables 2 and 3 (monkeys) and Table 4 (apes and human). Animal's activity rhythms, diets and locomotor types are cited additionally in each table.

I. Comparisons of phylogenetical development

In order to compare the phylogenetical development of the inferior olivary nuclei among primates, the allometry of each principal (IOPr) and accessory (IOAc) nuclei volumes and body masses are shown in Figs. 1, 2, 3 and 4. The IOAc is the total volume of the dorsal accessory nucleus and the medial one. Each regression line, which is based upon the values of 17 prosimian species, was applied as a standard line to compare size differences among the species investigated. As well known, STEPHAN used the values of the basal insectivores as a standard regression line (STEPHAN, 1967). Because data about IOPr and IOAc of insectivores are not readily available, comparison among species, subfamilies and families were tentatively made by using the regression line based upon the 17 prosimian species investigated. Regarding the information on the nuclear complex (IOPr + IOAc) in Insectivora, see STEPHAN *et al.* (1991). Thereby, the average values of size indices of prosimian species are 1.00 in each nucleus. In Table 5, a tentative size indices of each family or subfamily are cited

Table 1. Body weight, volumes[1)] of the inferior olivary nuclei and eco-ethological characteristics[2)] in Scandentia and prosimians

	Specimens (n)	Body Weight (g)	Inf.Oliv. Principal (mm^3)	Inf.Oliv. Acc.Med. (mm^3)	Inf.Oliv. Acc.Dors. (mm^3)	Med.+ Dorsal (mm^3)	Eco-ethological Characteristics Activity Time	Diet	Locomotor Type[3)]
Tupaia glis	2	170	0.672	0.774	0.501	1.28	D	Ins.	SAQ
Urogale everetti	1	275	0.661	0.848	0.433	1.28	D	Ins.	SAQ
Microcebus murinus	2	54	0.380	0.282	0.370	0.652	N	Ins.	AQP
Cheirogaleus major	2	450	1.62	1.30	1.31	2.61	N	Fru.	AQP
Cheirogaleus med.	1	177	0.726	0.415	0.610	1.03	N	Fru.	AQP
Lemur fulvus	2	1400	4.46	3.37	3.36	6.73	D	Fol.	AQP
Varecia variegata	1	3000	5.19	7.03	4.95	12.0	D	Fol.	AQP
Lepilemur ruficaud	1	915	1.71	1.67	1.42	3.09	N	Fol.	VCL
Avahi laniger	1	860	2.22	1.71	1.52	3.23	N	Fol.	VCL
Propithecus verr.	2	3480	5.95	6.68	3.52	10.2	D	Fol.	VCL
Indri indri	2	6250	8.94	7.23	4.56	11.8	D	Fol.	VCL
Daubentonia madagas	1	2800	10.7	7.98	5.37	13.4	N	Ins.	AQP
Loris tardigradus	2	322	0.921	1.08	0.844	1.92	N	Ins.	SCQ
Nycticebus coucang	2	800	1.96	1.77	1.41	3.18	N	Ins.	SCQ
Perodicticus potto	2	1150	2.14	3.43	1.95	5.38	N	Fru.	SCQ
Galago senegal.	2	186	0.8000	1.08	0.717	1.80	N	Fru.	VCL
Otolemur crassicaud.	2	850	1.35	2.43	1.47	3.90	N	Fru.	VCL
Galagoides demidoff	2	81	0.436	0.717	0.507	1.22	N	Ins.	VCL
Tarsius spp.	3	125	0.582	1.17	0.551	1.72	N	Ins.	VCL

1) Significant figures are presented in each Table.
2) Abbreviations: D–diurnal, N–nocturnal, Ins–insectivores, Fru–frugivores, Fol–folivores, AQP–arboreal quadrupedal prosimians, SAQ–semiarboreal quadrupeds, SCQ–slow-climbing quadrupedal prosimians, VCL–Vertical clinging and leaping prosimians.
3) After NAPIER and NAPIER (1967).

in average.

The principal nucleus (IOPr): Tupaiidae are clearly situated in a lower position than prosimians [Fig. 1, size index: 0.82, Table 5 (3)]. Among prosimians, the values of *Lemur* (No. 1 in Fig. 1, size index: 1.4) and *Daubentonia* (No. 2 in Fig. 1, size index: 2.1) are remarkably higher than those of other prosimian species. The *Daubentonia* have the highest values among prosimians and is in the range of the anthropoids distribution.

All species of anthropoids show a marked development in the IOPr, except for the marmosetts, of which, values are at the same level as those of prosimians (size index: 0.97). The position of *Cebus* (Fig. 2, No. 3, size index: 3.4) is the highest among the monkeys. *Hylobates* also situates in the highest range with that of human (Fig. 2, size index: 5.7 in *Hylobates;* 4.0 in human).

Concerning the ratios of the IOPr volumes compared to the volumes of the medulla oblongata (Table 5), a similar pattern with cases

Table 2. Body weight, volumes of the inferior olivary nuclei and eco-ethological characteristics[1)] in New World monkeys

	Specimens (n)	Body Weight (g)	Inf.Oliv. Principal (mm^3)	Inf.Oliv. Acc.Med. (mm^3)	Inf.Oliv. Acc.Dors. (mm^3)	Med. + Dorsal (mm^3)	Eco-ethological Characteristics Activity Time	Diet	Locomotor Type[2)]
Callithrix jacchus	2	280	0.767	1.13	0.864	1.99	D	Fru.	NAQ
Cebuella pygmaea	2	120	0.716	0.852	0.563	1.42	D	Fru.	NAQ
Saguinus midas	2	340	1.12	1.59	0.960	2.55	D	Fru.	NAQ
Saguinus oedipus	2	380	1.19	1.29	1.10	2.39	D	Fru.	NAQ
Cebus albifrons	1	3100	18.5	7.81	8.41	16.2	D	Fru.	NAQ
Aotus trivirgatus	2	830	2.20	1.92	1.26	3.18	N	Fru.	NAQ
Callicebus moloch	2	900	2.07	2.56	1.57	4.13	D	Fru.	NAQ
Saimiri sciureus	2	660	2.40	2.82	1.86	4.68	D	Fru.	NAQ
Pithecia monachus	2	1500	7.14	3.71	2.32	6.03	D	Fru.	NAQ
Alouatta seniculus	2	6400	18.1	7.80	5.39	13.2	D	Fol.	NSB
Ateles geoffroyi	1	8000	24.3	4.52	4.62	9.14	D	Fru.	NSB
Lagothrix lagothricha	2	5200	21.7	5.42	3.65	9.07	D	Fru.	NSB

1) Abbreviations: NAQ–New World arboreal quadrupedal simians, NSB–New World semibrachiators. For others, see Table 1.
2) After NAPIER and NAPIER (1967)

Table 3. Body weight, volumes of the inferior olivary nuclei and eco-ethological characteristics[1)] in Old World monkeys

	Specimens (n)	Body Weight (g)	Inf.Oliv. Principal (mm^3)	Inf.Oliv. Acc.Med. (mm^3)	Inf.Oliv. Acc.Dors. (mm^3)	Med. + Dorsal (mm^3)	Eco-ethological Characteristics Activity Time	Diet	Locomotor Type[2)]
Macaca mulatta	2	7800	26.6	8.73	6.93	15.7	D	Fru.	OTQ
Cercocebus albigena	2	7900	20.2	9.51	5.62	15.1	D	Fru.	OAQ
Papio anubis	2	25000	26.8	16.1	9.25	25.4	D	Fru.	OTQ
Cercopithecus ascanius	2	3400	15.0	6.53	4.38	10.9	D	Fru.	OAQ
Cercopithecus mitis	1	6300	13.2	7.41	3.48	10.9	D	Fru.	OAQ
Miopithecus talapoin	2	1200	9.24	4.36	3.11	7.47	D	Fru.	OAQ
Erythrocebus patas	2	7800	14.7	7.10	5.50	12.6	D	Fru.	OTQ
Colobus badius	2	7000	18.0	6.43	3.47	9.90	D	Fol.	OSB
Pygathrix nemaeus	1	7500	20.6	7.34	4.37	11.7	D	Fol.	OSB
Nasalis larvatus	1	14000	33.0	10.8	6.65	17.5	D	Fol.	OSB

1) Abbreviations; OTQ–Old World terrestrial quadrupedal simians, OAQ–Old World arboreal quadrupedal simians, OSB–Old World semibrachiators. For others, see Table 1.
2) After NAPIER and NAPIER (1967).

Table 4. Body weight, volumes of the inferior olivary nuclei and eco-ethological characteristics[1)] in apes and man

	Specimens (n)	Body Weight (g)	Inf.Oliv. Principal (mm^3)	Inf.Oliv. Acc.Med. (mm^3)	Inf.Oliv. Acc.Dors. (mm^3)	Med.+ Dorsal (mm^3)	Eco-ethological Characteristics: Activity Time	Diet	Locomotor Type[2)]
Hylobates lar	1	5700	47.5	13.6	5.92	19.5	D	Fru.	OTB
Pan troglodytes	1	46000	65.1	3.69	2.16	5.85	D	Fru.	OMB
Gorilla gorilla	2	105000	94.3	12.8	5.52	18.3	D	Fol.	OMB
Homo sap. sapiens	2	65000	189	14.5	8.50	23.0	D	Omni.	SBP

1) Abbreviations: Omni–omnivores, OTB–Old World typical brachiators, OMB–Old World modified brachiators, SBP–specialized bipedal hominids. For others, see Table 1.
2) After NAPIER and NAPIER (1967).

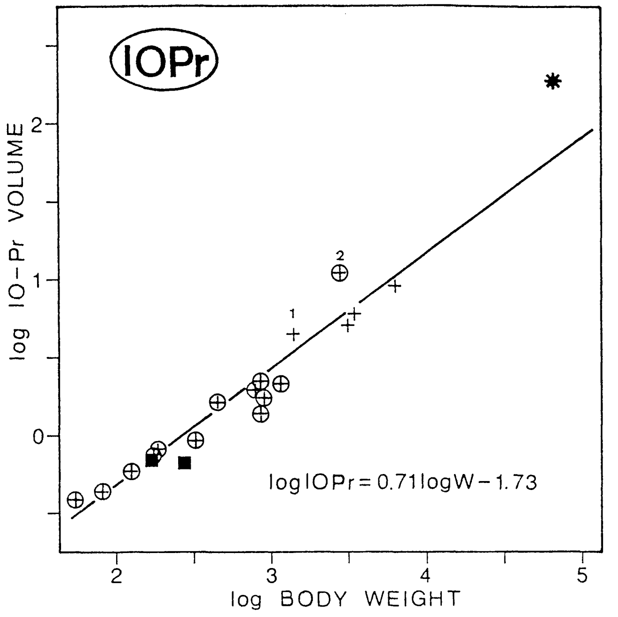

Fig. 1. The inferior olivary principal nucleus volumes (IOPr,mm^3) plotted against body weight (W, g) on a double-logarithmic scale. The reference line, which is determined by least-square regression analysis, is placed through the average of prosimians. Squares: Scandentia. Crosses: prosimians. Round: nocturnal. 1: *Lemur,* 2: *Daubentonia.* Star: human.

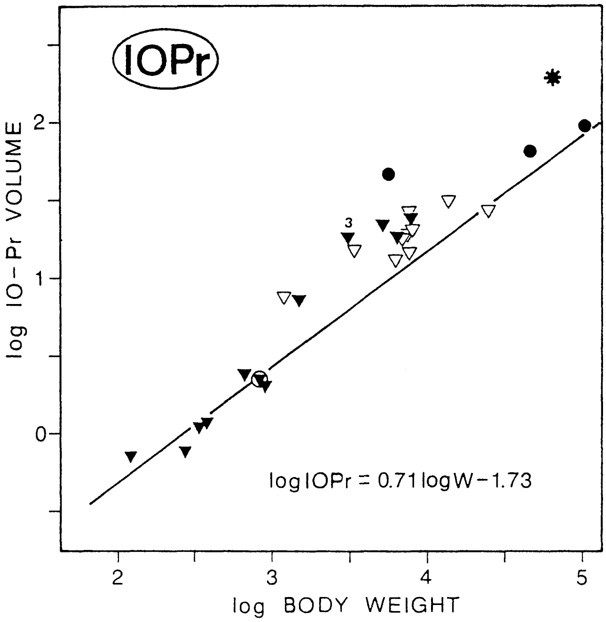

Fig. 2. This figure is made by the same method as Fig. 1. Black triangles: New World monkeys. Open triangles: Old World monkeys. Black rounds: apes. Star: human. 3: *Cebus.*

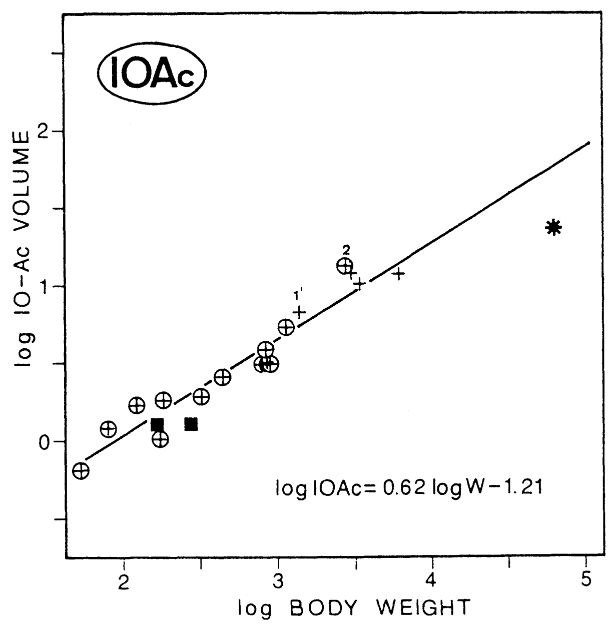

Fig. 3. The inferior olivary accessory nucleus volumes (IOAc, mm^3) plotted against body weight (W, g) on a double-logarithmic scale. The reference line, which is determined by least-square regression analysis, is placed through the average of prosimians. Squares: Scandentia. Crosses: prosimians. Rounds: nocturnal. 1': *Lemur.* 2': *Daubentonia.* Star: human.

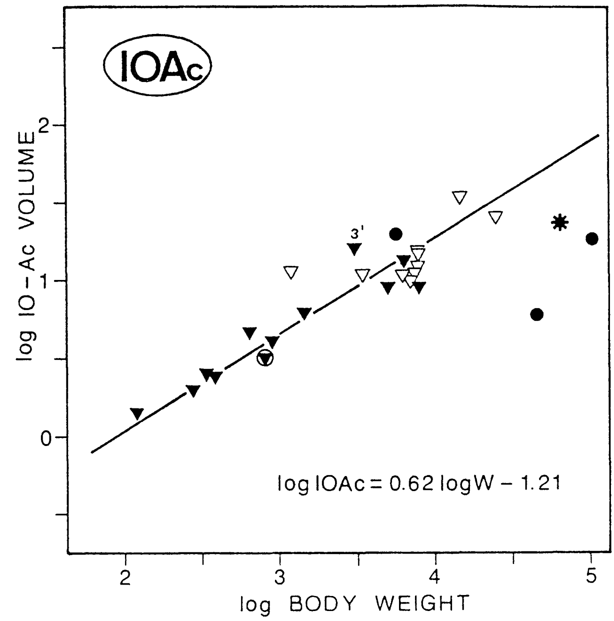

Fig. 4. This figure is made by the same method as Fig. 3. Black triangles: New World monkeys. Open triangles: Old World monkeys. Black rounds: apes. Star: Human. 3': *Cebus.*

of size indices is recognized as follows: 1) Remarkable volume ratios of the IOPr to the medulla oblongata among prosimians is found in aye-aye [67 in Table 5 (2)]. 2) Values in Callitrichidae are very low (29) and are under the prosimian average (40). 3) Values of *Hylobates* (210) belong to the highest group with those of human (200). Two species of great apes (*Gorilla* and *Pan*) show very high values as compared with the average monkey level (120 in the former *vs.* 72 in the latter). This is considered to be a matter of course. In size indices, however, they are low as compared with the average values of Old World monkeys, presumably because great apes (especially *Gorilla*) have large body size.

The accessory nuclei (IOAc): Among prosimians, the IOAc of *Daubentonia* shows a remarkable development, as is the case of IOPr [No. 2 in Fig. 3, size index: 1.6, Table 5 (3)]. But in general, all of the species were found to be distributed closely to the regression line for prosimians.

Development of the IOAc in monkeys (Fig. 4) is largely overlapping those of prosimians. Though their distribution is somewhat scattered, the average size indices in Table 5 show 1.00 in New World monkeys and 0.91 in Old World monkeys. The situations of two great apes and human are conspicuously low (Fig. 4), but the position *Hylobates* occupies is relatively high (See also size indices in Table 5).

The values of IOAc ratios in the medulla oblongata are extremely low in great apes and human, as is the case of size indices (Table 5).

Summary: Species demonstrating a relatively good development in both IOPr and IOAc are *Lemur* and *Daubentonia* in prosimians, and *Cebus* and *Hylobates* in anthropoids. In contrast to a great development of IOPr, the species having a poor IOAc are *Ateles* (size index: 2.3

Table 5. Relative development[1)] of each nucleus in primates: Each nucleus volume × 100/total nuclei volume is shown in the left column (1); each nucleus volume × 10,000/volume of medulla oblongata in the middle column (2); size indices, *i.e.* the ratio of nucleus volume observed to the volume predicted based upon the regression line of 17 prosimian species, in the right one (3)

	Specimens (n)	Principal inferior olivary N.			Accessory inferior olivary N.		
		(1)	(2)	(3)	(1)	(2)	(3)
Tupaiidae	2	34	25	0.82	66	47	0.75
Cheirogaleidae	3	39	39	1.1	61	61	0.84
Lemurini	2	35	43	1.2	65	80	1.3
Lepilemurini	1	36	38	0.74	64	69	0.73
Indriidae	3	40	53	1.0	60	78	0.90
Daubentoniidae	1	45	67	2.1	55	88	1.6
Lorisinae	3	33	36	0.85	67	74	0.93
Galaginae	3	28	27	0.91	72	72	1.1
Tarsiidae	1	28	28	1.0	72	84	1.4
Prosimians		35	40	1.0	65	76	1.0
Callitrichidae	4	32	29	0.97	68	62	1.0
Cebidae	8	54	77	2.0	46	61	1.0
N. W. monkeys		46	61	1.6	54	62	1.0
Cercopithecinae	7	56	80	2.0	46	61	0.98
Colobinae	3	65	98	2.0	35	54	0.73
O. W. monkeys		59	85	2.0	41	59	0.91
Hylobates lar		71	210	5.7	29	87	1.5
Pan troglodytes		92	110	1.8	8	24	0.12
Gorilla gorilla		84	130	1.4	16	10	0.23
Homo sapiens		89	200	4.0	11	24	0.39

Figures in italics represent an average for each taxon

in IOPr and 0.56 in IOAc; the percentage ratios of IOPr volumes per total nuclei ones: 73%), apes and human (Table 5). By observation under a microscope, a great development of IOPr in *Ateles* gives almost the same impression as those of apes. As a result of phylogenetic development of the inferior olivary nuclei of primates, we can recognize that the IOPr is clearly progressive from prosimians to human, although some overlaps exist among the range of each family, and subfamily (Fig. 2, Table 5). On the other hand, the IOAc is independent of this progressive trend. In the IOAc, there are found remarkable overlaps of values among species, subfamilies and families. The development of IOAc in great apes and human is far under the level of the prosimian average (Fig. 4, Table 5).

II. Relation with motor behaviors

Prosimians: Among three locomotor types of prosimians (AQP, VCL, SCQ in Table 1), the ranges of distribution in the development of IOPr and of IOAc overlap each other. But a relatively good development is recognizable in the AQP type in comparison with the other two types. Average size indices in IOPr are 1.3 in AQP, 0.94 in VCL and 0.85 in SCQ. In IOAc these are 1.1 in AQP, 1.0 in VCL and 0.93 in SCQ. Among species with the same type of locomotion, differences between species are also found. For example, in VCL group, *Galagoides* is 1.05 in size index of IOPr and 1.30 in that of IOAc, but *Lepilemur* is 0.74 in IOPr and 0.73 in IOAc. As cited by STEPHAN *et al.* (1988) from WALKER's paper (1979), *Galagoides* shows the most versatile type of locomotion, and the animals progress by running, walking, climbing quadrupedally, and leaping between supports, whereas *Lepilemur* shows the most typical VCL. Therefore, the remarkable difference of size indices between two species are considered to reflect their locomotion pattern.

In relation to classification of the dietary habits, difference of development of IOPr and IOAc are not found among three types of diets (Ins., Fru., Fol. in Table 1). For example, the IOPr develops remarkably in both *Lemur* and *Daubentonia,* but the former is frugivorous and the latter is insectivorous.

Regarding the extremely high development of IOPr and IOAc in *Daubentonia,* STEPHAN's opinion (1988) that this animal shows a secondary reduction in body mass (dwarfism), may be taken into consideration. *Daubentonia* is a highly adaptable form found in numerous habitats but never in high densities (TATTERSALL, 1982, cited by STEPHAN *et al.*, 1988). Indeed, the animal shows the great skill in using the skeletal finger to capture larvae of wood-boring insects.

Anthropoids: According to ERIKSON's classification (1963) of locomotor type, the arboreal quadrupedalism in New World monkeys (NAQ in Table 2) are divided into two types, the springer and the climber. Both IOPr and IOAc of the climber type clearly shows good development in comparison with the springer type. In size indices, the IOPr is 2.3 in the former and 0.97 in the latter, and the IOAc is 1.4 in the former and 0.99 in the latter. *Cebus,* as already described, has good development in both IOPr (3.4 in size index) and IOAc (1.8 in that) (Figs. 2 and 4, No. 3 and 3'). The animal is extremely active and agile, with considerable manupulative skill in handling objects (NAPIER and NAPIER, 1967).

The semibrachiation type (in both NSB and OSB in Tables 2 and 3) in monkeys is poorly developed in the IOAc in comparison with a good development of the IOPr (size index: 2.4 in NSB-IOPr, 2.0 in OSB-IOPr, 0.74 in NSB-IOAc, 0.73 in OSB-IOAc). In contrast, the arboreal quadrupedalism type (in both NAQ and OAQ in Tables 2 and 3) shows relatively a good development of IOAc (size index: 1.1 in NAQ, 1.3 in OAQ). Additionally, the OTQ type (Table 3) is clearly lower in development of both IOPr and IOAc than OAQ type (size index: 1.7 in IOPr in the former and 2.2 in the latter, 0.85 in IOAc in the former and 1.3 in the latter). These results are considered to reflect the locomotive characteristics of each type, though the nervous mechanism in the brain is yet unknown in detail. With all monkeys, the types of NAQ and OAQ develop maintaining balance between the IOPr and the IOAc. This fact seems to indicate that arboreal monkeys usually live in complicated, discontinuous, three dimensional space, as representated by AQ type, and need exceptional capacity in each pattern of locomotion and positional behavior.

Furthermore, our recent study showed a clear-cut differences between *Ateles* and *Macaca* in the kinesiological characteristics during vertical climbing (HIRASAKI *et al.*, 1992). The Ateles-type of climbing seemed to have more potential to develop into bipedalism than that of Macaca-type and the limb support patterns indicated a somewhat deliberate mode in *Ateles.* Concerning the volumetric comparisons on the IOPr, IOAc and the medial (CerM) and lateral (CerL) cerebellar nuclei (MATANO *et al.*, 1985a) in *Ateles* and *Macaca,* the following results are

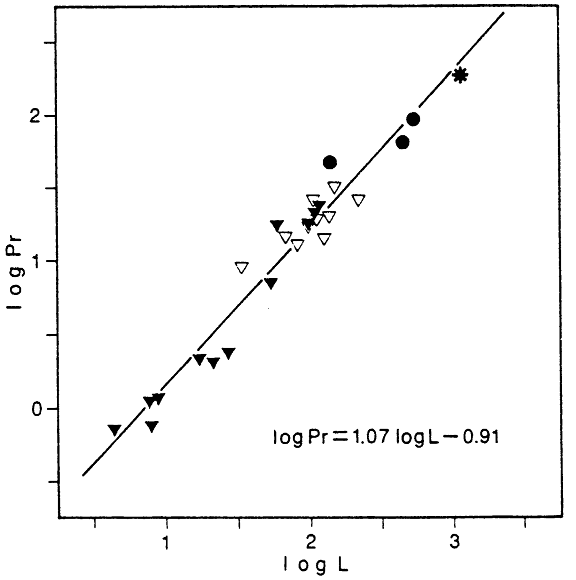

Fig. 5. The correlation between the lateral cerebellar nucleus volumes (mm^3, log L) and the inferior olivary principal nucleus volumes (mm^3, log Pr).

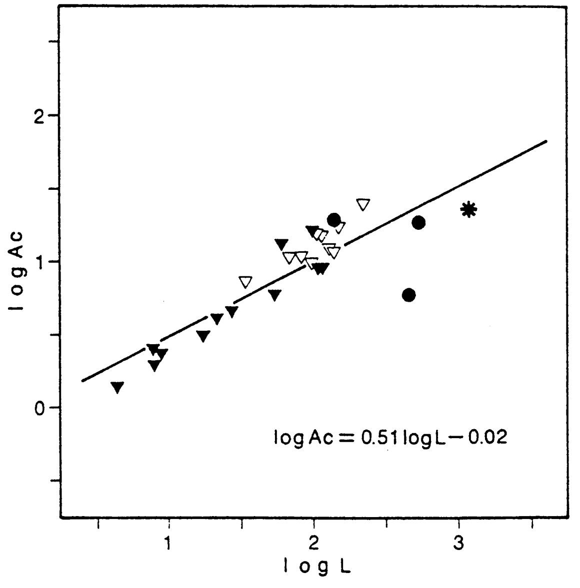

Fig. 6. The correlation between the lateral cerebellar nucleus volumes (mm^3, log L) and the inferior olivary accessory nucleus volumes (mm^3, log Ac).

Table 6. Correlation coefficients between size indices of ventral pons, three cerebellar nuclei, inferior olivary nuclei and vestibular nuclei in total species investigated

	Ventral pons	Medial cerebellar nucleus	Interpositus cerebellar nucleus	Lateral cerebellar nucleus	Inferior olivary principal nucleus	Inferior olivary accessory nucleus	Vestibular nuclear complex
Ventral Pons	–						
Medial cerebellar nucleus	–0.019	–					
Interpositus cerebellar nucleus	0.50	0.73	–				
Lateral cerebellar nucleus	0.91	0.15	0.61	–			
Inferior olivary principal nucleus	0.81	0.0097	0.48	0.85	–		
Inferior olivary accessory nucleus	–0.16	0.70	0.49	–0.098	0.044	–	
Vestibular nuclear complex	0.071	0.78	0.64	0.16	0.038	0.66	–

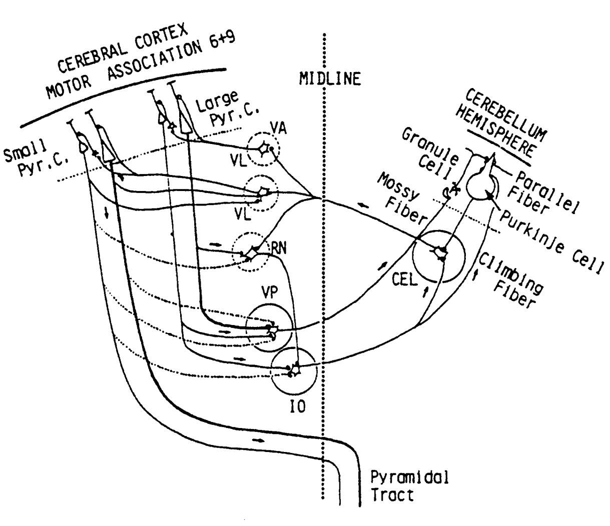

Fig. 7. Cerebro-cerebellar pathways linking association and motor cortices with the cerebellar hemisphere (After ECCLES, 1982 with a slight modification). VA, VL: Thalamic intermediated nuclei, RN: Red nucleus, VP: Ventral pons, IO: IOPr, CEL: Lateral cerebellar nucleus.

recognizable: The size indices of the IOAc and CerM were remarkably lower in *Ateles* (IOAc: 0.56, CerM: 0.71) than in *Macaca* (IOAc: 0.99, CerM: 1.1). On the other hand, the values of the IOPr and CerL indices were similar with these two species (IOPr: 2.3 and CerL: 1.8 in *Ateles;* IOPr: 2.5 and CerL: 1.7 in *Macaca*). The results of the comparison may reflect the specific mode of climbing in these two species, though the detailed neural mechanism is yet unknown.

The progressive development of IOPr in great apes and human is dependent upon their skillful voluntary hand movement rather than their locomotive characteristics.

III. Relations with other nuclei in motor system

When the results of IOPr are compared with those of three cerebellar nuclei (MATANO *et al.*, 1985a), the ventral pons (MATANO *et al.*, 1985b) and the vestibular nuclei (MATANO, 1986), the phylogenetical development of IOPr is recognized to be extremely similar with that of the lateral cerebellar nucleus. Both of them show a general evolutionary trend from prosimians to human progressively. On the other hand, the development of IOAc resemble closely that of the medial cerebellar nucleus. For example, Figs. 5 and 6 show the relation between the lateral cerebellar nucleus (L) and the IOPr (Pr) and between the L and the IOAc (Ac) among anthropoids, respectively. As shown in Fig. 5, Pr and L develop in a parallel fashion each other (the slope of regression line: 1.07), whereas Ac and L are independent of each other (the slope: 0.51).

Concerning gray matters investigated from prosimians to human, the correlation coefficients of the tentative size indices based upon 17 prosimian species are shown in Table 6. The IOPr shows a close relationship with the lateral cerebellar nucleus ($r = 0.85$), and the ventral pons ($r = 0.81$). On the other hand, the IOAc shows relatively high values between the medial cerebellar nucleus ($r = 0.70$) and between the vestibular nuclear complex ($r = 0.66$). Moreover, as shown in Table 6, the high correlation coefficients also exist between the ventral pons and the lateral cerebellar nucleus ($r = 0.91$), and the vestibular nuclear complex and the medial cerebellar nucleus ($r = 0.78$).

Putting these results together, we can easily conclude the following. First, the ventral pons, the lateral cerebellar nucleus and the IOPr are closely linked during the evolutionary process in the brain of prosimians to human progressively. These nuclei play an important role in cerebro-cerebellar hemisphere circuit for planning, programming and executing highly organized complex voluntary movements (Fig. 7 by ECCLES, 1982). Second, the importance of such voluntary movements increased in higher evolutionary levels and it is thereby a substantial component of general evolution. Third, on the other hand, the vestibular nuclear complex, the medial cerebellar nucleus and the IOAc have evolved to be closely related with each other and that is quite independent of a general evolutionary trend in primates. This is confirmed by the values of size indices in these nuclei of great apes and human, which are far inferior to the average prosimian. These nuclei are deeply related to muscle tonus, posture and simple, automatic movements (ECCLES, 1982). Thus, fourth, phylogenetic development of each nucleus in brain structure never yields with the same degree and shows a mozaic pattern. And it corresponds to evolution of habits and behaviors in animal life.

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抄　録

霊長類下オリーブ核の比較神経学的考察

保 野 彰 三

STEPHAN, H. の脳コレクションから，ツパイ2種，原猿17種，新世界サル12種，旧世界サル10種，テナガサル，ゴリラ，チンパンジーおよびヒトについて，下オリーブ核の容積を測定した．得られた結果から，アロメトリーによる系統発達を原猿を基準として考察した．

下オリーブ主核は原猿からヒトに至る前進的な進化傾向を示すのに対し，下オリーブ副核は霊長類の進化の一般傾向とは無関係であり，たとえば大型類人猿やヒトの副核の発達は，平均的原猿のレベルより低い．サル類のロコモータータイプからみると，樹上四足歩行型（AQP）が主核，副核ともに相対的にバランスよく発達し，半腕わたり型（NSB と OSB）や地上四足歩行型（OTQ）よりすぐれる傾向にあった．これまでに得られた橋底部，小脳核，前庭神経核の結果とも綜合して，小脳をめぐる運動系と関連させて討議した．

References

BARON, G., H.D. FRAHM and H. STEPHAN, 1988: Comparison of brain structure volumes in Insectivora and Primates, VIII. J. Hirnforsch., 29: 509-523.

CARPENTER, M.B., 1976: Human Neuroanatomy. Williams and Wilkins, Baltimore.

COURVILLE, J., C. de MONTIGNY and Y. LAMARRE, 1980: The Inferior Olivary Nucleus. Raven Press, New York.

ECCLES, J.C., 1982: The future of studies on the cerebellum. In: S.L. PALAY and V. CHAN-PALAY (eds.), The Cerebellum, New Vistas. Springer Verlag, Berlin, pp. 607-620.

ERIKSON, G.E., 1967: Brachiation in new world monkeys and in anthropoid apes. Symp. Zool. Soc. Lond., 10: 135-164.

GERHARD, L. and J. OLSZEWSKI, 1969: Medulla oblongata and pons. In: H. HOFER, A.H. SCHULTZ and D. STARCK (eds.), Primatologia, Bd. II, Teil 2, Lfg. 3, S. Karger, Basel.

HIRASAKI, E., S. MATANO, Y. NAKANO and H. ISHIDA, 1992: The vertical climbing in *Ateles geoffroyi* and *Macaca fuscata* with the comparative neurological background. In: S. MATANO, R. TUTTLE, H. ISHIDA and M. GOODMAN (eds.), Topics in Primatology, vol. III. Univ. Tokyo Press, Tokyo (in press).

KAPPERS, C.U.A., G.C. HUBER and E.C. CROSBY, 1960: The Comparative Anatomy of the Nervous System of Vertebrates, including Man, vol. III. Hafner, New York.

KUHLENBECK, H., 1975: The Central Nervous System of Vertebrates, vol. IV. S. Karger, Basel.

MATANO, S., 1986: A volumetric comparison of the vestibular nuclei in primates. Folia Primatol., 47: 189-203.

MATANO, S., G. BARON, H. STEPHAN and H.D. FRAHM, 1985a: Volume comparisons in the cerebellar complex of primates, II. Cerebellar nuclei. Folia Primatol., 44: 182-203.

MATANO, S., H. STEPHAN and G. BARON, 1985b: Volume comparisons in the cerebellar complex of primates, I. Ventral pons. Folia Primatol., 44: 171-181.

NAPIER, J.R. and P.N. NAPIER, 1967: A Handbook of Living Primates. Academic Press, London.

STEPHAN, H., 1967: Zur Entwicklungshöhe der Primaten nach Merkmalen des Gehirns. In: D. STARCK, R. SCHNEIDER and H.-J. KUHN (eds.), Neue Ergebnisse der Primatologie. Gustav Fischer Verlag, Stuttgart, pp. 108-119.

STEPHAN, H., G. BARON and H.D. FRAHM, 1981: New and revised data on volumes of brain structures in insectivores and primates. Folia Primatol., 35: 1-29.

STEPHAN, H., G. BARON and H.D. FRAHM, 1988: Comparative size of brains and brain components. In: H.D. STEKLIS and J. ERWIN (eds.), Comparative Primate Biology. vol. IV. Alan R. LISS Inc., New York, pp. 1-18.

STEPHAN, H., G. BARON and H.D. FRAHM, 1991: Insectivora. Springer Verlag, New York.

TATTERSALL, I., 1982: The Primates of Madagascar. Columbia Univ. Press, New York. (cited by STEPHAN *et al.*, 1988).

TILNEY, F., 1927: The brain stem of *Tarsius*. A critical comparison with other primates. J. Comp. Neurol.,

43: 371-432.
WALKER, A., 1979: Prosimian locomotor behavior. In: G.A. DOYLE and R.D. MARTIN (eds.), The Study of Prosimian Behavior. Academic Press, New York, pp. 543-565. (cited by STEPHAN *et al.*, 1988).

保 野 彰 三　大阪大学人間科学部　生物人類学教室
〒565　吹田市山田丘 1−2

Shozo MATANO　Department of Biological Anthropology
Faculty of Human Sciences, Osaka University
Yamadaoka 1-2, Suita-shi, Osaka 565, Japan